

STATEZERO LABS: MASTER PLATFORM SUMMARY

Date: June 2026

Purpose: Comprehensive prompt-ready project context document for ingestion by LLMs, Cursor Agents, or engineering teams to build out the StateZero web platform and computational engine.

Core Vision

StateZero Labs is a pre-seed TechBio startup focused on reprogramming fibrotic tissue (scarring) back to a healthy fetal blueprint using computational latent-space transcriptomics. The platform aims to treat severe skin trauma, deep dermal burns, and fibrosis without slow, traditional wet-lab trial and error.

1. Biological Framework & Core Pipeline Engine

The core software framework models biological networks as programmatic repositories to predict tissue state reprogramming vectors:

- **T_{pre} (Disease State):** Profiled from scRNA-seq datasets of hypertrophic scars, deep burns, or keloids.
- **T_{post} (Target Blueprint):** Profiled from healthy regenerative fetal skin or quiescent adult fibroblasts.
- **LINCS Integration:** Uses NIH LINCS Phase I L1000 dataset (GSE92742) signatures specifically filtered for `trt_lig` (Treatment: Ligand) to screen biological growth factors (e.g., EGF, FGF2, TGFB1) that minimize Mean Squared Error (MSE) to the blueprint.
- **Virtual CRISPR Module:** Simulates parallelized multi-gene knockouts targeting scale-free network master regulatory hubs (e.g., TGF- β /SMAD pathways) to collapse fibrotic trajectories.
- **Transcriptomic-to-Fingerprint Encoder (TFE):** Translates target genetic coordinates directly into Morgan Fingerprint chemical space to identify synthesizable candidate small molecules.
- **Lead Asset (SZ-01):** A computationally validated topical polypharmacology cocktail achieving an 89% in-silico cell-state reversal score.

2. Web Platform & UI Design Identity

The goal is a high-performance, ultramodern, minimalist web application constructed directly with code to avoid template restrictions and allow complete granular control over interactive pipeline visualizations.

2.1 Visual Aesthetic Guidelines

- **Theme:** Dark mode command-line interface (CLI) meets high-end bio-computation. Clean, stark layouts with extreme typography hierarchy.
- **Color Palette:** Base background #05050A / #06060F, primary surface card panels #141425, crisp text #F4F4FB. Interactive highlights use Neon Green (#22C55E) and Electric Violet (#7C3AED).
- **Visual Elements:** 1px subtle neon/glassmorphism borders, monospace text fragments for technical metrics, and cursor-responsive canvas particle backgrounds.

2.2 Required Site Architecture

1. **Hero Block:** Stark statement ("Reprogramming Tissue via Latent Space") backed by a subtle cursor-tracking grid or fluid WebGL simulation.
2. **Platform Architecture Grid:** A bento-box design depicting Level 1 (Orchestration/Nextflow), Level 2 (Core Virtual CRISPR & TFE Compute), and Level 3 (scRNA-seq Data Foundations).
3. **Dynamic Pipeline Tracker:** A custom interactive horizontal chart documenting asset SZ-01's progression from In-Silico validation through In-Vitro, Ex-Vivo, to Clinical phase trials.
4. **Company & Infrastructure:** Team section highlighting Pavel Abramau (CEO/Architect) and Prof. Paweł Sachadyn (Scientific Lead), linked with direct terminal-style access hooks to GitHub codebases and technical documentation data rooms.

3. Production-Ready Tech Stack Libraries

To implement custom pipeline animations and cursor interactions natively via code, the following tech stack must be deployed:

- **Framework:** Next.js (React) with Tailwind CSS for layout rendering.
- **Aceternity UI / Magic UI:** Pre-built, customizable copy-paste TypeScript/Tailwind components optimized for deep-tech styling (e.g., Background Beams, Tracing Beams, Neon Cards).
- **Framer Motion:** For robust, custom layout updates, page transitions, and programmatic scrolling triggers.
- **HTML5 Canvas / WebGL:** For custom performance-optimized interactive node networks without heavy asset overhead.

4. Development Environment & Next Steps

To avoid hardware limitations (e.g., local storage/RAM strain from massive 35GB GCTX datasets), the engineering workload uses a thin-client Mac connecting remotely over SSH to a 1TB high-capacity Ryzen processing workstation acting as the execution core.

Ingestion Protocol Guardrail

Ensure that dataset preparation scripts isolate GSE92742's Phase I ligand perturbations. Avoid GSE70138 Phase II matrices for growth factor routines, as they are strictly small-molecule (trt_cp) compound matrices.

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